# Week 11 Worksheet Solutions

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**A promise problem**: Consider a task where $f : \{0,1\}^n \to \{0,1\}$ is a boolean function. An algorithm queries $f(x), x \in \{0,1\}^n$ and must decide if $f$ is constant or balanced. We are promised that $f$ is either constant or balanced.

> (a) For $n = 1$, write the truth table for all the four possible functions $f(x), x \in \{0,1\}$. Show that two functions are constant, and two of these functions are balanced.

(a) *Answer: The truth table for constant function is given by* $f_1$ *and* $f_2$ *and the balanced functions is* $f_3$ *and* $f_4$*:*

| $x$ | $f_1$ | $f_2$ | $f_3$ | $f_4$ |
|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 |

> (b) For $n = 1$, how many queries does a deterministic, non-quantum algorithm needs to solve this problem?

(b) *Answer: 2 queries, since it has to check* $x = 0$ *and* $x = 1$ *case.*

> (c) For $n = 1$, would a *random* non-quantum algorithm perform better in this case? Why or why not?

(c) *Answer: No. A random quantum algorithm will use the first query as a reference bit. If the function is constant, then we will be right. If the function is balanced, there is a 50% chance we will be wrong on the second* *query. To reduce this error probability, we need more queries but then it's better to just use a deterministic algorithm.*

(d) For $n = 2$, write the truth table for two constant functions and six possible balanced functions $f(x), x \in \{0,1\}^2 = \{00, 01, 10, 11\}$.

(d) *Answer*:

| $x_1$ | $x_2$ | $c_1(x)$ | $c_2(x)$ | $b_1(x)$ | $b_2(x)$ | $b_3(x)$ | $b_4(x)$ | $b_5(x)$ | $b_6(x)$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |

Table 1: For $x = [x_1, x_2]$, the functions $c_1(x), c_2(x)$ are constant functions, $b_1(x), \ldots, b_6(x)$ are balanced functions.

(e) For $n = 2$, are there functions which are neither constant nor balanced? Give an example.

(e) *Answer: Yes, for* $n > 1$, *functions can be neither balanced or constant. Let* $f'$ *be a function that is neither balanced nor constant. A valid example of* $f'$ *is any function that yields 4 single-bit outputs with more than one unique value and that doesn't have an equal number of zeros or ones over the full input space is a valid answer, e.g.* $0, 0, 0, 1$ *or* $1, 0, 1, 1$. *The example* $0, 0, 0, 1$ *is stated formally below,*

$$f'(x) = \begin{cases} 0, & x = [0, 0], \\ 0, & x = [0, 1], \\ 0, & x = [1, 0], \\ 1, & x = [1, 1]. \end{cases} \tag{1}$$

(f) For $n > 1$, how many queries does a *deterministic* non-quantum algorithm needs to solve this problem? Explain why.

(f) *Answer: We have to check half the input space* $2^n/2 = 2^{n-1}$. *In the worst case, the function will be the same value for all of these inputs. We have to check one other input to see if the value is the same or different. If* *same, $f$ is constant else balanced. Total queries are $2^{n-1}+1$.*

(g) Will using a *random* non-quantum algorithm help? Explain why.

(g) *Answer: Yes, randomness helps for $n > 1$. Query one random input. This is your reference. The probability the next query will give the same output as the reference is $1/2$. The probability that next query will be the same as the second query is also $1/2$. The probability that all 3 queries give the same output as the reference bit is $(1/2)(1/2)$. After seeing $k-1$ queries with the same output as the reference bit, I state that function is constant. However if I see a different value in the next query, then the function is balanced and I would be wrong. The probability that I am wrong is $\left(\frac{1}{2}\right)^k$. This error probability decays rapidly with $k$, e.g. $0.5^{10} = 0.00098$.*